

each other tick their seconds away at different rates, although the effect would only be observable at extremely high relative speeds. This led Einstein to conclude that space and time are intrinsically linked in a “continuum.”

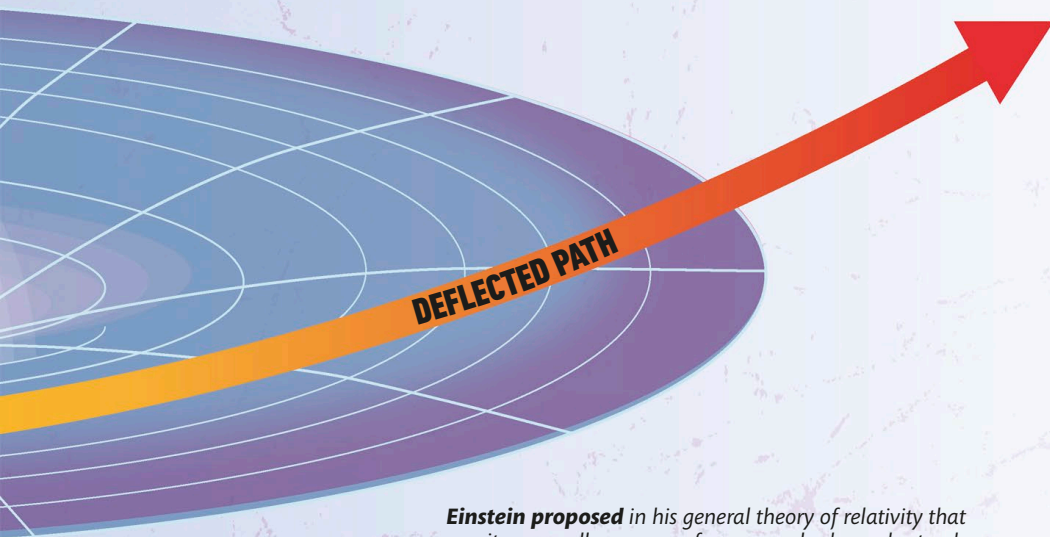
### Energy equals mass

Einstein's fourth paper was an extension of special relativity that proved that mass and energy are relative and are aspects of the same property. He expressed this as  $E = mc^2$ , which states that the energy (E) of an object is the same as its mass (m) multiplied by the square of the speed of light ( $c^2$ ). Since  $c^2$  is a very large number—about 90 million billion—it follows that even a small amount of mass can contain a huge quantity of energy. This is the basis of nuclear power: atomic nuclei are broken apart and lose mass, resulting in large amounts of energy.

Over the next 10 years, Einstein developed relativity to include gravity. Though Isaac Newton had first described gravity, he had been unable to explain its origins or the laws of physics that govern it. In his 1915 general theory of relativity, Einstein explained that time slows down in intense gravitational fields, which cause time, matter, and light to bend as they pass close to very massive objects, such as the Sun.

### Testing the theory

Einstein's predictions were put to the test in 1919, when physicists observing a total solar eclipse from several locations noted that stars appeared to be out of place due to the light they emitted being bent around the Sun. This shift made no sense under Newton's theory of gravity, but corresponded exactly with Einstein's general theory of relativity.



**Einstein proposed** in his general theory of relativity that gravity, normally seen as a force, can also be understood as a distortion of space-time caused by objects with large mass, like a large ball lying on a rubber sheet that causes a smaller object to roll toward it.

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